

Dashboard §11.4.170 host-render matrix expansion — Deployments / Fleet / Groups / Overview

Revision: 1 Last modified: 2026-07-09T00:00:00Z

0. Scope

This stream (Stream R, dashboard/ build+host-render ownership per §11.4.119) closes the last 4 owed rows of the dashboard §11.4.170 host-render sub-matrix documented in docs/research/frontend_production_readiness_20260709/READINESS.md §2:

```
Deployments / Fleet / Groups / Overview | ✖ | ✖ | Repointed  
to tokens but not yet host-render-proven ... Owed.
```

The 5 screens already proven (Login, AppShell, AuditScreen, ReleaseList, ArtifactUploadScreen) are untouched — their goldens, evidence files, and test bodies are unchanged except for one shared assertion fix described in §3.

1. What was added

- dashboard/hostrender/harness-main.tsx — mounts the 4 remaining real screens under ?screen=deployments|fleet|groups|overview:
 - DeploymentList (src/screens/DeploymentsScreen.tsx, route / deployments) — pure client-side screen, no /api call on mount (an “open by id” form only).
 - FleetHealth (src/screens/FleetScreen.tsx, route /fleet) — calls GET /telemetry/overview; stubbed with a canned TelemetryOverview (12 success / 2 failure / 3 installing events, by_state 4 buckets).

- `GroupList` (`src/screens/GroupsScreen.tsx`, route `/groups`) — calls `GET /groups`; stubbed with a 2-row canned `DeviceGroupList` (`canary-fleet`, `production-fleet`).
- `DashboardOverview` (`src/screens/OverviewScreen.tsx`, route `/`) — calls `GET /releases?limit=5` (reuses the existing `RELEASES` canned fixture, path-suffix matched) and `GET /healthz`; stubbed with `{status: "ok"}`.
- `window.fetch` stub extended with 3 new path matches: `/healthz`, `/telemetry/overview`, `/groups`.
- `dashboard/hostrender/screens.hostrender.spec.ts` — 4 new `ScreenSpec` entries (`deployments`, `fleet`, `groups`, `overview`), each driven through the IDENTICAL dual-oracle test battery already proven for the first 5 screens: golden image-diff, DOM-bounds/text layout oracle (self-validated), explicit pixelmatch analyzer (self-validated), committed-baseline-rejects-mutated, and light ↔ dark distinctness. New per-test captured evidence writes to a **separate** evidence directory (`docs/qa/20260709-dashboard-hostrender-matrix-expand/`, this directory) via a new `evidenceDir` field on `ScreenSpec` + an `evidenceDirFor()` helper — the prior round's `docs/qa/20260709-dashboard-vendoring-complete/` directory is untouched.
- 8 new committed golden PNGs under `dashboard/hostrender/screens.hostrender.spec.ts-snapshots/`: `deployments-{light,dark}-chromium-linux.png`, `fleet-{light,dark}-chromium-linux.png`, `groups-{light,dark}-chromium-linux.png`, `overview-{light,dark}-chromium-linux.png`.

2. Shared-oracle fix (§11.4.6 / §11.4.107(10) — genuine bug, not a bluff)

Minting the Fleet golden surfaced a real defect in the SHARED layout-oracle test body (not a bluff — the test correctly FAILED, it just failed for the wrong reason): the convention `requiredLabels[0]` MUST be the heading's own text broke for `FleetHealth`, whose `<h1>Fleet</h1>` text is a **substring** of the same page's Card title="`Fleet overview`". Hiding the `<h1>` (the canonical §11.4.170 regression) does not remove the substring "Fleet" from `document.body.innerText`, because "Fleet overview" is still rendered — so the original assertion (`bad.failures` must contain a message citing `requiredLabels[0]`) failed even though the oracle correctly detected the mutation via a DIFFERENT, always-present signal: element "heading" not rendered (no box / 0x0) (the literal `<h1>` DOM node collapses to a null bounding box the instant

display:none hits it — this fires unconditionally for every screen, independent of incidental text-substring overlaps elsewhere on the page).

Fix applied in `screens.hostrenderer.spec.ts` (one shared test body, applies to all 8 screens): the assertion now accepts EITHER the missing-label failure OR the heading-not-rendered failure as proof the heading’s disappearance was caught. This does **not** weaken the analyzer — `bad.pass === false` is still independently required, and the overlap failure is still independently required; it only broadens WHICH specific failure string counts as proof of heading-disappearance detection, using a more mechanically reliable per-element signal (the box collapsing to 0x0) alongside the pre-existing full-page-text signal. All 4 pre-existing screens still pass via the original text-substring signal (verified — see §4); Fleet now passes via the box-collapse signal. Re-ran with `--update-snapshots` then a clean pass after the fix — both green (§4).

3. Per-screen × theme verdict table (the 4 new screens)

All ratios below are pulled directly from the captured JSON evidence files in this directory (not restated from memory) — see the cited filename per cell.

Screen	Theme	Golden-good	Layout oracle good/bad	Image-diff self-check (good ↔ good / good ↔ mutated)	Baseline reject mutated
Deployments (DeploymentList)	light	PASS	PASS / FAIL-detected	0 / 0.5105 (image-diff-selfcheck-deployments-light.json)	dimsDiffer=false ratio=0.5178 (baseline-rejects-mutated-deployments-light.json)
Deployments (DeploymentList)	dark	PASS	PASS / FAIL-detected	0 / 0.5120 (image-diff-selfcheck-deployments-dark.json)	dimsDiffer=false ratio=0.5193 (baseline-rejects-mutated-

Screen	Theme	Golden-good	Layout oracle good/bad	Image-diff self-check (good ↔ good / good ↔ mutated)	Baseline reject mutated
					deployments-dark.json)
					dimsDiffer=true
Fleet (FleetHealth)	light	PASS	PASS / FAIL-detected	0 / 0.5088 (image-diff-selfcheck-fleet-light.json)	ratio=1 (baseline-rejects-mutated-fleet-light.json)
Fleet (FleetHealth)	dark	PASS	PASS / FAIL-detected	0 / 0.5230 (image-diff-selfcheck-fleet-dark.json)	dimsDiffer=true ratio=1 (baseline-rejects-mutated-fleet-dark.json)
Groups (GroupList)	light	PASS	PASS / FAIL-detected	0 / 0.5096 (image-diff-selfcheck-groups-light.json)	dimsDiffer=false ratio=0.5167 (baseline-rejects-mutated-groups-light.json)
Groups (GroupList)	dark	PASS	PASS / FAIL-detected	0 / 0.5119 (image-diff-selfcheck-groups-dark.json)	dimsDiffer=false ratio=0.5189 (baseline-rejects-mutated-groups-dark.json)
Overview (DashboardOverview)	light	PASS	PASS / FAIL-detected	0 / 0.5075 (image-diff-selfcheck-overview-light.json)	dimsDiffer=false ratio=0.5142 (baseline-rejects-mutated-overview-light.json)
Overview (DashboardOverview)	dark	PASS	PASS / FAIL-detected	0 / 0.5082 (image-diff-selfcheck-overview-dark.json)	dimsDiffer=false ratio=0.5147 (baseline-rejects-mutated-

Screen	Theme	Golden-good	Layout oracle good/bad	Image-diff self-check (good ↔ good / good ↔ mutated)	Baseline reject mutated
					overview-dark.json)

Every image-diff self-check verdict field reads "SELF-VALIDATED: analyzer passes golden-good AND flags golden-bad" (good ↔ good < 0.001, good ↔ mutated > 0.01 — actual observed good ↔ mutated ratios are ~0.51, i.e. roughly half the frame changed, driven by the injected 900×520 red overlay + the hidden heading). Every layout-oracle good run has `pass:true`, `failures:[]`; every bad run has `pass:false` with both a heading-disappearance failure (text-substring match for Deployments/Groups/Overview, box-collapse match for Fleet — see §2) and an overlap failure present.

Honest note on Fleet's `dimsDiffer=true`: the committed-baseline-rejects-mutated check found the mutated Fleet render has different pixel dimensions than the golden baseline (the injected fixed-position overlay plus the resulting layout changes push the captured root element's rendered height past the baseline's), so the check short-circuits to `ratio=1` per its own documented logic (`dimsDiffer ? 1 : 0`) rather than running `pixelmatch` on mismatched-size buffers. This is still a valid, mechanically correct GOLDEN-BAD rejection (a dimension change is itself proof the mutated render does not match the golden) — not a SKIP, not a bluff.

4. Build / test / e2e counts (before → after)

Gate	Before (prior round, per READINESS.md §1.2/§2)	After (this round)
<code>npm run build (tsc ×2 + vite build)</code>	exit 0	exit 0 (re-run this round, clean)
<code>npm run test:run (unit, vitest)</code>	107 passed (12 files)	107 passed (12 files) — unchanged, no <code>src/</code> edits
<code>npm run e2e:hostrender (--</code>	n/a	81 passed (9 login + 72 screens: 4 old screens unaffected 36

Gate	Before (prior round, per READINESS.md §1.2/§2)	After (this round)
update-snapshots, mint)		+ 4 new screens × 9 tests each = 36)
npm run e2e:hostrender (clean, no update)	45 passed (9 login + 36 dashboard screens)	81 passed (9 login + 72 dashboard-screen tests) — clean, no snapshot drift

Both e2e runs (mint + clean) captured in full; the clean run is the authoritative verdict (/tmp scratch logs, not committed — the committed evidence is the per-test JSON/PNG files enumerated in §3 plus the golden PNGs in dashboard/hostrender/screens.hostrender.spec.ts-snapshots/).

Test-count arithmetic for the screens spec: 8 screens × (4 per-theme tests × 2 themes + 1 distinctness test) = 8 × 9 = 72; + 9 login-spec tests = 81 total under playwright.hostrender.config.ts.

5. Screens NOT covered here (honest scope boundary)

- DeploymentCreateScreen, DeploymentDetail, DeviceDetail, GroupCreateScreen, GroupDetail — sub-screens (create/detail forms) of the 4 top-level nav items this task targeted. READINESS.md §2 names the 4 owed rows as Deployments / Fleet / Groups / Overview (the top-level list screens matching the AppShell nav + App.tsx route map), matching DeploymentList/FleetHealth/GroupList/DashboardOverview exactly as proven here. The detail/create sub-screens remain an honest, separately trackable gap — not silently implied proven (§11.4.3 SKIP-with-reason posture: not in this task's named scope, not claimed).
- No screen in this batch required a SKIP — all 4 rendered cleanly with the existing component-render-stub harness pattern (§11.4.27 permitted: real components, stubbed window.fetch + stubbed login, no view logic faked).

6. Updated sub-matrix (dashboard)

Screen	light	dark	Evidence
Login	✓	✓	docs/qa/ 20260709- dashboard- hostrender/
AppShell frame	✓	✓	docs/qa/ 20260709- dashboard- vendoring- complete/
AuditScreen	✓	✓	docs/qa/ 20260709- dashboard- vendoring- complete/
ReleaseList (releases)	✓	✓	docs/qa/ 20260709- dashboard- vendoring- complete/
ArtifactUploadScreen	✓	✓	docs/qa/ 20260709- dashboard- vendoring- complete/
Deployments (DeploymentList)	✓	✓	this directory (§3)
Fleet (FleetHealth)	✓	✓	this directory (§3)
Groups (GroupList)	✓	✓	this directory (§3)
Overview (Dashboard Overview)	✓	✓	this directory (§3)

dashboard host-render coverage = 9 screens × {light,dark} (was 5).
Every top-level nav screen the AppShell exposes (App.tsx route map) is now host-render proven; the READINESS.md §2 “owed” row for dashboard is closed.

7. Files changed / created (all under dashboard/ + this docs/qa/ dir)

- dashboard/hostrender/harness-main.tsx (edited — imports, canned data, fetch stub, ScreenHost switch)
- dashboard/hostrender/screens.hostrender.spec.ts (edited — EVIDENCE_DIR_EXPAND, evidenceDir field + evidenceDirFor(), 4 new ScreenSpec entries, shared layout-oracle assertion broadened per §2)
- dashboard/hostrender/screens.hostrender.spec.ts-snapshots/deployments-{light,dark}-chromium-linux.png (new)
- dashboard/hostrender/screens.hostrender.spec.ts-snapshots/fleet-{light,dark}-chromium-linux.png (new)
- dashboard/hostrender/screens.hostrender.spec.ts-snapshots/groups-{light,dark}-chromium-linux.png (new)
- dashboard/hostrender/screens.hostrender.spec.ts-snapshots/overview-{light,dark}-chromium-linux.png (new)
- docs/qa/20260709-dashboard-hostrender-matrix-expand/ (this directory — 52 captured evidence files: *-actual.png, diff-bad-*.png, layout-oracle-{good,bad}-*.json, image-diff-selfcheck-*.json, baseline-rejects-mutated-*.json, light-vs-dark-distinctness-*.json, this EVIDENCE.md)

No files under clients/ota-manager/, server/, or design-systems/ were touched. No git commands were run by this stream (per task instruction) — the new baseline PNGs + evidence files are left in the working tree for the conductor to review and commit.